

# Analyzing Systems Using Data Dictionaries

CSE 4407

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# The Data Dictionary

- What

- Think of it as a specialized dictionary, but for *data*
- A reference work of **metadata**
- **Example:** Customer ID, Order Status Code, Product Price, etc.
- “The system’s official glossary!”

- Core Purpose

- Guides analysts during analysis and design
- **Collects, coordinates, and confirms data term meanings** across the organization
- Ensures everyone speaks the same “data language”

# The Need for a Data Dictionary

- Automated Tools?

- Many DBMS and CASE tools include automated DDs
- They catalog items, provide templates, ensure uniformity

- Crucial Role: Keeping data CLEAN!

- Ensures data consistency → Avoids chaos!
- Example: External Customer stored as “Ext”, “E”, or “1”? → Bad!
- “A DD is like a data ‘style guide’ - keeps things tidy!”

- Vital for Analyst

- Even with tools, *you* need to know:
  - What goes into a DD?
  - Conventions and structure
  - How it's developed → Often starting from DFDs!
- Helps conceptualize the system deeply

# Key Uses of a Data Dictionary

- **Validate DFDs:** Ensure completeness and accuracy → Does every flow/store/process have defined data?
- **Develop Screens and Reports:** Provides a starting point for fields needed
- **Define Data Stores:** Determines the exact contents of files/tables
- **Develop Process Logic:** Helps create detailed logic (mini-specs) for DFD processes
- **Create XML:** Structures data definitions for data exchange

# Data Repository vs. Data Dictionary

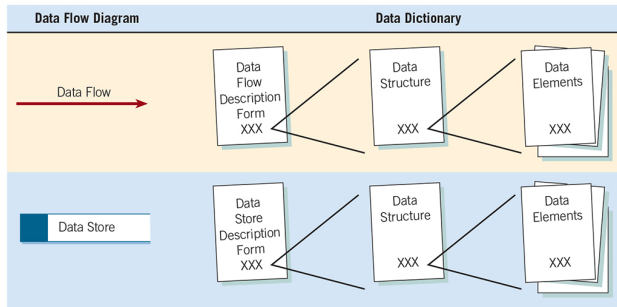
- **Data Dictionary:** Focuses on data definitions and procedures
- **Data Repository:** A *larger* collection of project information → Project Encyclopedia
- **Inside a Repository**
  - Procedural Logic and Use Cases
  - Screen and Report Designs
  - Data Relationships
  - Project Requirements and Deliverables
  - Project Management Info
- Central hub for everything

# How DD Entries are Born

- Creating the Dictionary:  
Examine DFDs!

- Steps

- Identify a **Data Flow** or **Data Store** on the DFD
- Create a basic Description Form for it
- Define its Data Structure
- Define each individual Data Element



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Demo: **Customer Order Confirmation Example**

# Data Flow Definition: The System's Conversations

- **Data Flows:** Represent data moving in the system (inputs, outputs, internal)
- **Starting Point:** Define **system inputs/outputs first**
- **How:** Interviews, Observation, Analyzing Documents/Existing Systems
- **Key Info to Capture (per flow)**
  - Unique Name
  - Source and Destination
  - Type
  - Associated Data Structure(s)
  - Volume and Comments

| Data Flow Description                                                                                                                                                         |                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| ID                                                                                                                                                                            |                                                                                                                              |
| Name                                                                                                                                                                          | Customer Order                                                                                                               |
| Description                                                                                                                                                                   | Contains customer order information and is used to update the customer master and item files and to produce an order record. |
|                                                                                                                                                                               |                                                                                                                              |
| Source                                                                                                                                                                        | Destination                                                                                                                  |
| Customer                                                                                                                                                                      | Process 1                                                                                                                    |
| Type of Data Flow                                                                                                                                                             |                                                                                                                              |
| <input type="checkbox"/> File <input checked="" type="checkbox"/> Screen <input type="checkbox"/> Report <input type="checkbox"/> Form <input type="checkbox"/> Internal      |                                                                                                                              |
| Data Structure Traveling with the Flow                                                                                                                                        | Volume/Time                                                                                                                  |
| Order Information                                                                                                                                                             | 10/hour                                                                                                                      |
| Comments                                                                                                                                                                      |                                                                                                                              |
| Order record information for one customer order. The order may be received by Web entry, email, FAX, or by the customer telephoning the order-processing department directly. |                                                                                                                              |
|                                                                                                                                                                               |                                                                                                                              |
|                                                                                                                                                                               |                                                                                                                              |
|                                                                                                                                                                               |                                                                                                                              |

# Data Structures: The Blueprint

- **Data Structure:** Defines *what* elements make up a data flow or data store
- **Method:** Algebraic Notation
  - = means “is composed of”
  - + means “and”
  - { } means repetition
  - [ ] means either/or
  - ( ) means optional
- **Structured Records:** Group related elements
- **Goal:** Break down structures into their basic Data Elements

Demo: [Data Structure Description Example](#)



# Data Structures: Logical vs. Physical

- Two Views

- Logical

- The *user's view or business perspective*
    - What data does the business need? → Customer Name, Item Price
    - Reflects user's mental model

- Physical

- The *implementation view*
    - Includes elements needed for the *system to work efficiently*  
→ Keys, Status Codes, Passwords

Customer Billing  
Statement

= Current Date +  
Customer Number +  
Customer Name +  
 $\sum_1^5 \{\text{Order Line}\} +$   
(Previous Payment  
Amount) +  
Total Amount Owed +  
(Comment)

Order Line

= Order Number +  
Order Date +  
Order Total

# Data Elements: Foundation

- **What:** The **smallest meaningful unit of data** → Cannot be broken down further
- **Define ONCE:** Each element gets a **single, authoritative definition in the DD**
- **Key Characteristics**
  - Element Name (Unique, descriptive)
  - Aliases (Synonyms used elsewhere)
  - Description (Clear meaning)
  - Length and Data Type
  - Validation Rules (How to ensure it is correct)
  - Base or Derived?

Demo: [Element Description Form Example](#)

# Data Elements: Core Attributes

- **Length:** How much space does it need?
  - **Numeric**
    - Based on largest possible value
    - Room for growth
  - **Text**
    - Use standards/sampling (Book)
    - Trade-offs if too short
- **Type:** What kind of data is it?
  - Numeric, Character/Varchar, Date, Currency, Binary, etc.
  - Choice depends on usage (display, calculation)
- **Format:** How is it displayed/entered?
  - **Uses symbols:** 9 for number, X for any character, Z for zero suppress, and , . / - for editing
  - **Example:** 999-99-9999 for Social Security Number in US

# Data Elements: Advanced Characteristics

- **Validation Criteria:** Ensuring data accuracy
  - **Discrete**
    - **Fixed values** (e.g., Color Codes: 'BL', 'WH', 'GR')
    - Use **List or Table**
  - **Continues**
    - **Range of values** (e.g., GPA: 0.00 - 4.00)
    - **Use Range or Limits**
  - Check **Digits** (for key fields)
- **Base vs. Derived**
  - **Base**
    - Entered directly (e.g., Customer Name, Price)
    - Must be stored
  - **Derived**
    - **Calculated by a process** (e.g., Order Total, Age)
    - **May or may not be stored**
- **Default Value:** Pre-filled value to speed up entry (e.g., Default State)
- **Aliases:** Listing synonyms (e.g., Customer Number = Client Number)

# Data Stores: Where Data Lives

- What
  - Where the system's data resides persistently
  - Files, databases, tables
  - Group related elements into **structural records** (often based on entities like Customer, Product, Order)
  - May need to **combine info from *multiple* data flows**
- Purpose: Stores collections of base elements (and sometimes derived)
- Key Info
  - ID and Name
  - File Type and Format
  - Size/Growth
  - Primary and Secondary Keys

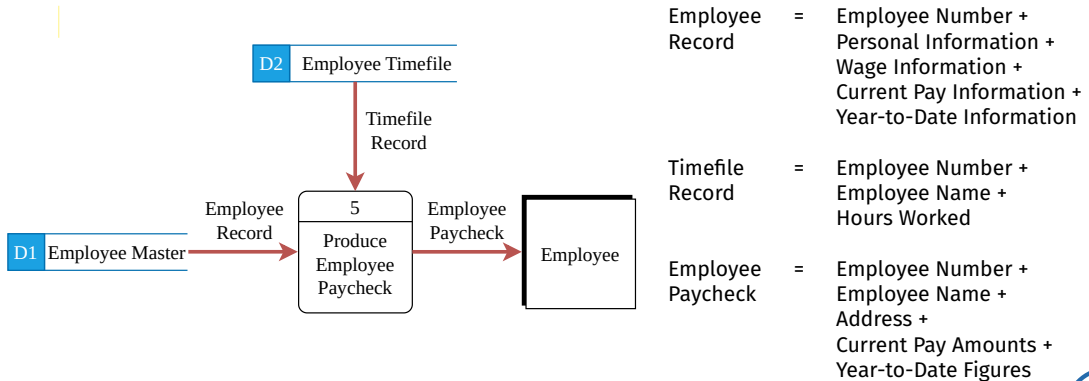
Demo: [Data Store Description Form Example](#)

# Creating and Data Dictionary: When and How

- Flexible Timing
  - Option 1: Create *after* DFDs are finalized
  - Option 2: Create *concurrently* with DFDs (Top-Down)
- Top-Down Approach (Recommended)
  - Initial Interviews → Sketch **Diagram 0**
  - Create initial DD entries → **High-level Flows and Structures**
  - More Analysis (Interviews, Observations, Docs) → **Refine DFDs (Child Diagrams)**
  - Expand **DD with detailed Structures and Elements**
- Analogy: Like designing a building!

# DFD Levels and Data Dictionary Consistency

- **Crucial:** DD detail **MUST** match the DFD level it describes
- **Balancing Act**
  - Any data structure or element flowing into or out of a child process...
  - ...**MUST** be contained **within a data flow shown on the parent diagram** for that process



# Step 1: Analyzing Inputs and Outputs

- Systematically identify and categorize ALL system inputs and outputs
- **Why:** Often define user interactions and core data requirements
- **Tool:** Input/Output Analysis Form
- **Key Info**
  - Descriptive Name
  - User Contact
  - Input or Output?
  - Format
  - Sequencing Elements
  - Element List

Demo: [Input and Output Analysis Form Example](#)



## Step 2: Identifying Structures from I/O Analysis

- Mine the I/O Forms for Structures!
- Examine the element list closely
- Look for
  - Elements that logically belong together → First Name, Last Name, Middle Initial
  - Repeating elements or groups → Multiple order lines
  - Optional elements → Apartment Number
  - Mutually exclusive elements → Different payment type details
- Action: Group related elements into **Structural Records** in DD

## Step 3: Developing Data Stores

- **Focus:** Data at rest
- **Purpose:** Hold permanent or semi-permanent information needed beyond a single transaction
- **How to choose**
  - Analyze processes
  - What info is temporary?
  - What needs to be kept long-term?
- **What to store**
  - Primarily **store base elements**
  - Derived data?
    - Calculated values *don't have to be* stored; can often be recalculated
    - Storing them is a design choice → Storage space vs. calculation time

Demo: **Data Store Example**

# Data Stores vs. User Views

- **Data Stores**
  - Aim to store data efficiently, often normalized, representing core entities
  - Data at rest
- **User Views**
  - Structures designed specifically for how a user wants to see data on a particular report on screen
  - Data presentation
- **Relationship:** Might pull data from multiple data stores and present it in a specific, combined format
- **Analogy**
  - Data Stores = A library with books organized by genre, author, and title.
  - User View = A custom reading list you've created for a weekend getaway, which includes books from various genres that fit the mood you want.

# Data Dictionary: The Payoff!

- Make DD Work for You
  - Ideal DD: Automated, Interactive, Online, Evolutionary
  - NOT just a document, a living tool
- Golden Rule: KEEP IT CURRENT!
  - Must be updated whenever the system design changes
  - Outdated DD is worse than useless → Misleading
- Perspective: View DD creation as an activity that parallels analysis and design, not a separate, final task

# Use Case 1: Creating Screens, Reports, and Forms

- **Definition to Design:** DD is a direct input for UI/Report design
- **How**
  - DD provides the elements needed and their lengths
  - Arrange these elements logically on the screen/page using design principles
- **Mapping Structures to Layout**
  - **Structural Records:** Group related fields together visually
  - **Repeating Groups:** Often become columns in a table or list on the layout

Demo: Order Picking Slip Data Structure vs. Order Picking Slip

# Use Cases 2 and 3: Generating Code and Analyzing Design

- **User Case 2: Generating Code Structures**

- DD Definitions can directly generate:
  - Database table schemas
  - Program code structures
- Ensures code matches the defined data structures

- **Use Case 3: Analyzing System Design:** Use DD to find logical flaws in your DFDs

- **Base Elements Rule:** Base elements appearing in an output flow *must* have been input to that process
- **Derived Elements Rule:** Derived elements *must* be created by a process
- **Data Store Rule:** Elements flowing into/out of a data store *must* exist within that data store's definition in the DD

## Use Case 4: Creating XML

- **What:** eXtensible Markup Language
  - The Universal Data Translator
  - **Primary Use:** Exchanging data between businesses, systems, or software
- **Key Concept:** Content vs. Format
  - HTML → Defines how data looks like
  - XML → Defines what the data is
  - **Note:** Presentation is handled separately
- **Benefits**
  - Define data once, transform for multiple outputs
  - Select and share only necessary data

# Benefits of a Well-Used Data Dictionary

- **Saves Time:** Reduces ambiguity and rework, especially if started early
- **"Single Source of Truth":** Central repository for all data definitions
- **Resolves Disputes:** Provides clear, agreed-upon meanings for data terms
- **Aids Maintenance:** Invaluable reference for understanding unfamiliar or legacy systems
- **Foundation for Automation:** Enables automated documentation, code generation, and validation checks